

Exercise Activity – Fall 2025 Midterm (10/8/2025)**You can tear the front page, but be sure to write your name/ID on the second page.**

First Last Name: _____

ID: _____

Lab Section: _____

- 4 total questions will be given.
- Lowest score dropped: Only your best 3 questions will count toward the grade.
- The last question is graded as all-or-nothing (either full points or zero).
- Maximum total = 110%.
- The pseudocode section is worth up to 10% per question.
- Code comments are worth 5% — just include a few key logic explanations.

Q1. Taylor Series Approximate $\sin(x)$ and $\cos(x)$ using Taylor series:

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

$$\sin(z) = z - \frac{1}{3!}z^3 + \frac{1}{5!}z^5 - \frac{1}{7!}z^7 + \frac{1}{9!}z^9 \pm \dots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} z^{2n+1}$$

$$\cos(z) = 1 - \frac{1}{2!}z^2 + \frac{1}{4!}z^4 - \frac{1}{6!}z^6 + \frac{1}{8!}z^8 - \frac{1}{10!}z^{10} \pm \dots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} z^{2n}$$

Tasks

- 1) Read x (radians) and error tolerance ϵ (>0).
- 2) Compute $\sin(x)$ and $\cos(x)$ by adding terms until the next term's absolute value $< \epsilon$.
- 3) Print the number of terms used for each, the approximations, and compare them with math library values.
- 4) If $|\cos(x)|$ is not too small, print $\tan(x) \approx \sin(x)/\cos(x)$ and the library $\tan(x)$.

Rules

- Use loops and helper functions (e.g., factorial or iterative term update).
- Stop early when precision ϵ is met.

Test Case 1 (Input): $x = 1.5708$ ($\approx \pi/2$), $\epsilon = 1e-6$ *Output:* $\sin \approx 0.999999$ (terms ≈ 6), lib=1.000000 $\cos \approx 0.000002$ (terms ≈ 5), lib ≈ 0.000000

tan: large (division by tiny cos)

Test Case 2 (Input): $x = 3.1416$ ($\approx \pi$), $\epsilon = 1e-5$ *Output:* $\sin \approx -0.000002$ (terms ≈ 7), lib=0.000000 $\cos \approx -1.000003$ (terms ≈ 8), lib=-1.000000 $\tan \approx 0.000002$, lib ≈ 0.000000

Q2: Bitwise Operations – Permissions and Fields

A 16-bit unsigned integer stores three fields:

- Bits 0–3: permissions (bit0=read, bit1=write, bit2=exec, bit3=admin)
- Bits 4–7: group code (0–15)
- Bits 8–15: user code (0–255)

Tasks

1. Extract each field using bitwise operators (&, |, ^, ~, <<, >>).
2. Write a function to set or clear a permission bit k (0–3).
3. Write a function to update the group code to a new 4-bit value v (0–15).
4. Print all fields before and after updates.

*Test Cases Case 1:**Input:*

value = 0xABCD

set k=1(k is position) → 1 (set write)

set group → 5

Output:

Before: perm=0xD group=0xC user=0xAB

After : perm=0xF group=0x5 user=0xAB

*Test Cases Case 2:**Input:*

value = 0x000F

clear k=3 (admin)

set group → 0xF

Output:

Before: perm=0xF group=0x0 user=0

After : perm=0x7 group=0xF user=0

Q3 Write a C program to merge two 1D arrays into a third array that includes duplicates, **ordered by the number of duplicates**. If two numbers have the same count, sort them in ascending order.(33.33 points)

Input example:

First array: 7 7 7 6 6 5 4

Second array: 2 2 2 3 3 4 5 5 5 5 6 7

Expected output:

Merged array: [5, 7, 2, 6, 3, 4]

Explanation:

5 appears 5 times

7 appears 4 times

2 appears 3 times

6 appears 3 times

3 appears 2 times

4 appears 1 time

Q4: Write a C program that manages a simple list of student records using arrays. Each record has three fields: Name, ID, and Section.

Your program should:

1. Read initial student records from a text file (input.txt). Each line has the format:
Name,ID,Section
 2. Process commands from the same file:
 - add Name ID Section → Add a new student.
 - del ID → Delete the student with that ID.
 3. After finishing, sort the records by Name (alphabetical order).
 4. Print the final list in the format:
 5. End programming by typing “end”.
- Name, ID, Section**

Example Test Case

Input file (input.txt):

Lily,12345,L01
Bob,99123,L02
Carol,55555,L01

del 99123

Expected Output:

Carol,55555,L01
Lily,12345,L01

add Dave 22222 L03

Expected Output:

Carol,55555,L01
Dave,22222,L03
Lily,12345,L01

add Bob 99123 L02

Expected Output:

Bob,99123,L02
Carol,55555,L01
Dave,22222,L03
Lily,12345,L01

